ALGORITHMS III

PSEUDOCODE COMMAND SET • THREE WAYS TO WRITE AN ALGORITHM

YEAR 10 • UNIT 1: PROBLEM SOLVING

WEEK 2 | CONTINUATION LESSON

TODAY'S OBJECTIVES

Move beyond the starter examples from the previous lessons.

- Use the Pearson command set for input, output, assignment, selection and repetition.
- Use an array, a counter, arithmetic and a decision in one complete solution.
- Show the same algorithm as a written description, a flowchart and pseudocode.
- Check that all three representations express the same logic.

SUCCESS CHECK	I can explain what happens, draw the control flow, and write command-set syntax that matches it.

ONE SOLUTION • THREE VIEWS

Quick recall: the representation changes; the algorithm's logic should not.

WRITTEN DESCRIPTION

Numbered, unambiguous steps in ordinary language.

Best for: explaining the plan clearly before coding.

FLOWCHART

Standard symbols connected by arrows.

Best for: seeing order, decisions and repeated paths at a glance.

PSEUDOCODE

Structured statements from the Pearson command set.

Best for: writing logic precisely without choosing a programming language.

KEEP THE LOGIC CONSISTENT

same inputs → same processing → same decisions → same outputs

COMMAND SET: MOVE DATA

Three familiar commands become the building blocks of a complete solution.

INPUT
PROCESS
OUTPUT

The data path

`KEYBOARD → SET → DISPLAY`

RECEIVE • INPUT

```
RECEIVE PagesToday FROM (INTEGER) KEYBOARD
```

SET • ASSIGN OR CALCULATE

```
SET Total TO Total + PagesToday
```

SEND • OUTPUT

```
SEND Total TO DISPLAY
```

INTEGER = whole number • REAL = decimal • BOOLEAN = True/False • CHARACTER = one character

OPERATORS + DECISIONS

Choose a condition, then make the two possible paths explicit.

OPERATOR GROUP	SYMBOLS FROM THE SPEC
ARITHMETIC	+ - * / ^ MOD DIV
	MOD = remainder • DIV = integer division • ^ = exponent
RELATIONAL	= <> > >= < <=
	<> = not equal • >= / <= include equality
LOGICAL	AND OR NOT
	AND = both true • OR = either true • NOT reverses

Example condition: `Total >= 100`

IF / ELSE / END IF

```
IF Total >= 100 THEN
    SET Status TO 'Challenge achieved'
ELSE
    SET Status TO 'Keep reading'
END IF
```

IF the expression is true, the first branch runs. Otherwise, the ELSE branch runs.

REPETITION: CHOOSE THE LOOP THAT FITS

The command set provides several ways to repeat a command.

FOR … TO … DO

Known number of repeats.

```
FOR Day FROM 0 TO 4 DO
```

FOR … STEP …

Count by a chosen interval.

```
FOR I FROM 0 TO 100 STEP 10 DO
```

REPEAT n TIMES

Known number of repeats.

```
REPEAT 5 TIMES
```

WHILE … DO

Pre-conditioned: may run zero times.

```
WHILE Total < 100 DO
```

REPEAT … UNTIL

Post-conditioned: runs at least once.

```
REPEAT … UNTIL Total >= 100
```

FOR EACH

One pass for each array element.

```
FOR EACH Word FROM WordsArray DO
```

Use END FOR / END REPEAT / END WHILE / END FOREACH as shown in the command set.

ARRAYS + OTHER COMMAND-SET TOOLS

Reference: specification PDF pages 43–49. Not every tool is needed in today's model.

ARRAYS + STRINGS

- 1D: Array[index]. 2D: Array[row, column]. Indices start at 0.
- `&` appends; STRING & non-STRING converts to STRING.

IDENTIFIERS + CONST

- Are names for containers that must be descriptive.
- CONST Example:

```
CONST REAL PI;
SET PI TO 3.14159.
```

BUILT-IN FUNCTIONS

`LENGTH()` works on arrays/strings.
`RANDOM(n)` generates a number from 0 to n.

COMMENTS + DEVICES

- `#` begins a comment.
- `KEYBOARD` / `DISPLAY` are common devices; others are context-specific.

FILE HANDLING

`READ File Record` reads a record.
`WRITE File Record` writes a record.

SUBPROGRAMS

`PROCEDURE` / `FUNCTION` define reusable commands. A function may `RETURN` a value; `Name(...)` calls it.

Notation: <...> means 'replace with a value'; do not type the brackets. Type coercion is automatic when needed (INTEGER + REAL → REAL).

WORKED TASK: THE 5-DAY READING CHALLENGE

One problem. We will express it in all three forms.

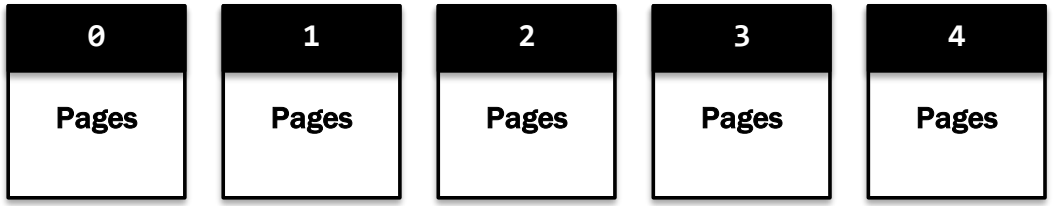
THE TASK

A student records the number of pages read on five days. The algorithm must store each day's input, calculate the total and average, then report whether the 100-page challenge was achieved.

SUCCESS RULE

```
Total pages >= 100
```

FIVE ARRAY POSITIONS



index starts at 0

Total = sum of all five entries

Average = Total / 5

MODEL 1: WRITTEN DESCRIPTION

Keep the steps precise, ordered and complete—especially where the process repeats.

1. Create a five-position Pages array and set Total to zero.
2. For each day from 0 to 4, ask the student how many pages they read.
3. Store that input in the matching Pages position and add it to Total.
4. After all five days, calculate Average by dividing Total by 5.
5. If Total is at least 100, set Status to ‘Challenge achieved’; otherwise set it to ‘Keep reading’.
6. Display Total, Average and Status.

Check: does step 5 describe both outcomes? Does step 3 happen once for every input?